

Weeks 2-10

Facilitator guides

Preparation, pacing, hints, proofs, source notes, and extra-page solutions.

Grade bands are entry points. Begin with objects, then try to explain what happens.

Give one page at a time. The packet is a menu, not a checklist for one hour.

Keep facilitator solutions separate from student pages.

Week and investigation	PDF pages
02 Switches and lamps	2-5
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Print US Letter, single-sided, at 100% / Actual Size.

Page ranges above count this cover as page 1. Bookmarks open each week.

Weekly footers retain their own page numbers and activity IDs for the use log.

Revised September 20, 2026. Planned activities; record actual use after teaching.

Prepare the toggle lab

The mathematical question is reachability, then optimization: which lamp patterns can local pair-flips produce, and how cheaply? This replaces the old missing-stroke core. Do not mix the old union rule with toggling.

Materials / seven children / two adults

Bring **29 two-sided counters**: four per K-1/middle child (24), five for upper; 8 spares; pencils and scratch paper. Paper circles with one shaded face work. Tap edges with fingers; no electronics. Print 3 K-1, 3 middle, 1 upper, 1 extra reserve, and this guide. Lend a spare for the six-ring.

Readiness and entry points

K-1: no independent reading; match two counter faces; count or pair up to four. Adult reads and models both endpoints flipping, including ON to OFF. Reasoning: undoing and an invariant.

2-3: labels 1-4, counting to four, odd/even or visible pairing; read-aloud works. Reasoning: classify cases, organize all possibilities, explain interior cancellation.

4-5: count to six, track yes/no choices, compare lengths. Reasoning: necessity versus sufficiency, exhaustive two-branch argument, construction plus lower bound.

Extra: read a graph, parity, follow paths; no algebra notation required. The abstraction is one theorem covering all finite graphs, then a unique solution on trees.

A shared 60-minute hour

0-10: explore counters and make pictures freely. **10-15**: demonstrate one edge press and its undo. Launch: “Can we make any picture we want?” **15-35**: K pair targets; middle target collection; upper cheapest solutions. Adult A anchors K; organizer visits middle and upper. **35-40**: movement/reset: stand, stretch, leave work ready to return to. **40-55**: continue the collection or use the optional proof paths below; offer the six-ring/extra only if wanted. **55-60**: share one discovery or impossibility; tidy counters. Give one page at a time. Rotate mover/target-maker/checker in triplets, each retaining a mat and counters. Schedule an adult visit to hear the upper child’s optimality argument.

Hints that leave room to discover

Check that ON flips OFF. Ask what a repeated press does. For parity, compare both OFF/both ON/mixed endpoints; for paths, count touches at a middle vertex. Cover later table columns and decide one edge at a time. A stalled child can stay on four lamps.

Satisfying stops / optional proof

K-1 stop: make and undo two pair targets. **Optional**: explain the impossible singleton by pairing ON lamps.

2-3 stop: collect eight four-lamp targets. **Optional**: prove odd targets fail and paths make every pair.

4-5 stop: solve both five-ring targets minimally; explain why one press fails. **Optional**: show each first-edge choice forces the rest, then use complements to prove the two-press bound.

Parity and all the pictures

K-1 solutions and explanations

Two identical presses undo each other; replaying any list in reverse undoes it. Top/right (1,2) needs 12. Top/bottom (1,3) uses 12 then 23. A single ON lamp is impossible under the pair rule. Both OFF endpoints become ON (+2), both ON become OFF (-2), or one ON trades places (0). Starting at zero, the number ON stays even. Children may demonstrate these three cases and pair the counters instead of saying “parity.” The new one-lamp button breaks this invariant and makes the singleton immediately.

Middle: the complete set

The eight states, listed by their ON lamps, are:

Number ON	Possible sets
0	none
2	12, 13, 14, 23, 24, 34
4	1234

Here “13” in the *state list* means lamps 1 and 3, not a legal edge. Avoid conflating a target list with a move list. All six pairs can be made by a path on the square. Each interior vertex is touched twice and returns OFF; each end is touched once. To make all four, use edges 12 and 34. No odd-size state is possible. This proves both that the list is complete and that every listed state is reachable. There are 12 blank recording boxes so their count does not give away the answer.

Why paths generalize

Pair the desired ON vertices. For each pair, choose a path and press its edges. At an intermediate vertex, contributions cancel in pairs; endpoints give one flip. Shared edges or vertices cause no problem: all flip counts add modulo two. An edge pressed twice can be deleted from the list. On any connected graph, exactly the even targets are reachable. Connectivity matters; see the extra.

What is being added?

Old shape addition used union: a shared stroke was kept once. Toggling is **symmetric difference**: something present twice cancels. This new operation restores information in the identity $(A \triangle B) \triangle B = A$. A one-minute stick-picture comparison is an optional bridge for returners, not a second core activity. Lowell Handout 11, problem 11.3, asked for splitting and recombining shapes; Handout 9, problem 9.1, used door toggles indexed by divisors. Here the adjacent-pair move and reachability problem are new.

Two answers, then an optimum

Upper page 1: constructions plus lower bounds

Target $\{1,3\}$: edges $\{12,23\}$ or $\{51,45,34\}$. Minimum **2**. One press can light only adjacent lamps; these are not adjacent. Target $\{1,2,3,4\}$: $\{12,34\}$ or $\{23,45,51\}$. Minimum **2**, because one press changes at most two lamps. Children should supply both the successful move list and the reason one press cannot suffice.

Upper page 2: exactly two reduced solutions

For target $\{1,3\}$, the decision table is:

	51	12	23	34	45	lamp 5
First no	no	yes	yes	no	no	OFF
First yes	yes	no	no	yes	yes	OFF

Once 51 is chosen, lamp 1 forces 12, lamp 2 forces 23, and so on. There can be at most two solutions using each edge zero or one times. Complementing an edge set preserves the target, because each lamp has exactly two incident edges: a flip count of 0, 1, 2 becomes 2, 1, 0, with the same parity. Thus every reachable target has exactly two reduced solutions. There are infinitely many unreduced move lists if repeated presses are permitted; state the distinction. For target $\{1\}$, first-no forces only 12,23,34,45, leaving lamp 5 ON incorrectly; first-yes forces only 51, again leaving lamp 5 ON. Both branches fail. Parity explains the same obstruction globally.

Upper page 3: the five-ring speed limit

An even target is reachable by paths. Its two reduced press sets have sizes k and $5 - k$; at least one is at most 2. Any repeated presses can be cancelled without changing the target, so the smaller reduced set is globally shortest. On an odd ring the two sizes never tie. On the six-ring, opposite lamps 1 and 4 have solutions $\{12,23,34\}$ and $\{61,56,45\}$, each size 3. The same forced-choice proof shows there is no third reduced set, hence no shorter route. More generally the two cycle solutions have sizes $k, n - k$.

Undergraduate interpretation

Write OFF=0 and ON=1. Each edge is a column with two 1's in a vertex-edge incidence matrix. A move set solves $Ax = t$ over the two-element field. The all-edge cycle is a nonzero null vector; on a single cycle it is the only one. Our forced-choice proof is binary elimination without matrix notation. The connected-graph image is the even-parity subspace; its dimension is $n - 1$.

Networks and real optimization

Extra: every component matters

Target $\{A,D\}$ is impossible: the triangle and the separate edge each contain an odd number of target vertices, while every move changes two vertices in one component. Target $\{A,C,D,E\}$ uses AC and DE, minimally two presses. In general, a target is reachable *if and only if each component contains an even number of targets*. Necessity is component parity; sufficiency pairs targets within each component and uses paths. Isolated vertices are components too.

Extra: why a tree has one solution

Delete an edge e . Count desired ON vertices on either side. Presses wholly on one side affect its parity by zero; the only move crossing the cut is e . Thus e is pressed an odd number of times exactly when that side has an odd target count; otherwise it is pressed an even number of times (possibly zero). The two sides agree because the whole target is even. These decisions uniquely specify every edge. Existence follows from the path construction. They therefore give the unique reduced solution and a shortest move list; repeated edges can only lengthen it. A leaf-removal algorithm gives another constructive explanation.

Research connection and its boundary

The pressed-edge set has odd degree exactly at the target vertices T : this is a **T-join**. The children's optimization is exactly minimum-cardinality T-join on a cycle. General networks can have many cycle choices; edge costs lead to the minimum-weight T-join problem and route-inspection algorithms. We prove the elementary parity and tree/cycle cases here, not the general matching algorithm or its polyhedral theorem. **Edmonds and Johnson (1973)**, *Matching, Euler tours and the Chinese postman*, *Mathematical Programming* 5, 88–124. The publisher abstract identifies matching, Euler tours, and the optimization result: doi:10.1007/BF01580113.

Cornuéjols, *Combinatorial Optimization: Packing and Covering*, Chapter 2, pp. 27–29, defines T-joins, explains shortest-path/matching reduction, and states the Edmonds–Johnson theorem: author's monograph. It uses acyclic representatives; a cycle can be deleted without changing odd-degree vertices and without increasing nonnegative cost.

Undergraduate anchor: MIT 18.06SC, Graphs, Networks, Incidence Matrices. Its usual real-valued matrix framework motivates the representation; the binary adaptation and proof above are ours.

Teaching sources and use history

Math Circle by the Bay, preface pp. viii–x, advocates themes with deep mathematical context, manipulatives, varied pace, and reserve challenges. Rozhkovskaya, Lesson 3 “At the lesson,” item 1, lets children attempt and explain before a table is introduced; Lesson 7, item 1, warns how missed moves can distort a game investigation. Our staffing and timing are adaptations. Record actual tasks as F02-K/M/U/X-v2; preparation is not evidence of teaching. Reserve the general graph version for a later year if unused. **Verification:** `verify.py` enumerates move subsets for 4-, 5-, and 6-cycles, checks two complementary solutions per even target, minima, and the extra examples. The arguments above stand independently of computation.

Codes become permutations

The main question is structural: how long can a reversible finite machine take to reset? Encoding is the entry point; loop decomposition supplies the explanation. These are deterministic repeated keys, not random swaps.

Materials and preparation

Make two circle/triangle/square sets per K-1 child (18 slips), one A-D set per middle child (12), and A-E plus F-H for upper/extra. Hand-lettered scraps suffice. Bring **13 counters**: three per K-1 child and one per older child; pencils and paper. Print 3 K-1, 3 middle, 1 upper, and 1 extra reserve. Read aloud; letters are labels, not spelling tasks.

Prerequisites

K-1: recognize three shapes, follow one arrow, count to three with support; reason about reversibility and repeat. **2-3**: recognize A-D and count to six; no multiplication needed. Track a consistent mapping and classify loops. **4-5**: count to six, coordinate multiple cycles; reason that an exhaustive classification establishes a maximum. Multiples/LCM can grow from lists of return turns. **Extra**: list partitions by largest part, find common multiples up to 15; build an upper-bound argument. Factorials are context, not required arithmetic.

A shared 60-minute hour

0-10: explore symbol cards and invent a short key. **10-15**: show the three-shape cycle. Launch: “What happens if we code the code again?” **15-35**: K follows shape keys; middle compares P/Q; upper tests the five-symbol key. Adult A anchors K; organizer visits middle and upper. **35-40**: movement/reset: stand, stretch, leave work ready to return to. **40-55**: continue with loop classification or an optional proof below; composition and the eight-letter extra are alternatives. **55-60**: share a machine and its return time; tidy cards. Give one page at a time. Upper page 3 is an alternative direction, not required before the extra. Rotate encoder/decoder/checker. An adult should hear the upper child’s loop argument and offer a counterproposal.

Helpful questions

“Where does this letter go next?” “Is every letter home?” “Can two arrows enter one symbol?” If enumeration stalls, choose the largest loop, then split what remains. Walk counters around loops and list their return turns.

Satisfying stops / optional proof

K-1 stop: return all shapes after three turns; undo a message. **Optional**: explain why merging two inputs prevents unique decoding.

2-3 stop: compare P/Q and build a four-turn key. **Optional**: use all five loop partitions to rule out six turns.

4-5 stop: build the 3+2 key and explain its first return at six. **Optional**: cover the supplied table, classify by largest loop, and prove six is the maximum.

Reversibility and loops

K-1 checked results

The three-shape key sends circle-triangle-circle to triangle-square-triangle. The input giving square is triangle. Every symbol returns after three turns; the order of names or the starting point does not affect that count. Under the changed key, circle and triangle swap while square stays put. All are home after two turns; square is also home after one. A key sending both circle and triangle to square loses information: the receiver cannot determine which was sent.

Middle page 1

P sends ABBA to **BCCB**; decoding CAAC gives **BCCB**. A one-to-one key preserves equality and inequality between positions, so ABBA cannot become ABCD. Its first and last outputs must match, as must the middle pair.

Turn	P	Q
0	ABCD	ABCD
1	BCAD	BADC
2	CABD	ABCD
3	ABCD	BADC
4	BCAD	ABCD

P has a three-cycle A-B-C-A and fixed D, so order 3. Q has two swaps A-B and C-D, so order 2. A single fixed letter does not certify that the whole key has returned.

Middle page 2: complete classification

A four-cycle A-B-C-D-A has order 4. The possible cycle-size partitions and their orders are:

Loop sizes	4	3+1	2+2	2+1+1	1+1+1+1
Order	4	3	2	2	1

These exhaust four letters by sorting on the largest loop. Order 6 is impossible with four letters; separate loops of size 3 and 2 require five. Counting all possible keys gives $4 \cdot 3 \cdot 2 \cdot 1 = 24$, but the classification uses five cases. Do not replace the investigation with a factorial exercise.

Why every reversible key splits into loops

Follow arrows from any symbol. Finiteness forces a repeated symbol. The first repeat must be the starting symbol: if a later point were first repeated, it would have two distinct predecessors, violating the one-to-one rule. Remove that loop and repeat with an unused symbol. Loops are disjoint and exhaust all symbols. On a loop of length a , the first return is a and all returns occur at multiples of a . All loops return together at their least common multiple. This is a child-accessible proof, not just a pattern from examples.

A maximum and noncommuting keys

Upper: the five-letter example

Successive messages are ABCDE, BCAED, CABDE, ABCED, BCAED, CABED, ABCDE. The 3-cycle returns on turns 3,6,9,...; the 2-cycle on 2,4,6,... The first common return is **6**. A five-cycle takes only 5, so the longest individual loop need not be optimal.

Sizes	5	4+1	3+2	3+1+1	2+2+1	2+1+1+1	1+1+1+1+1
Order	5	4	6	3	2	2	1

Cover the supplied table before children classify. Largest loop 5 or 4 leaves just one split; largest 3 leaves 2 or 1+1; largest 2 leaves 2+1 or 1+1+1; largest 1 forces five singletons. Ask why these exhaust the leftovers. This complete list plus the loop theorem proves no five-letter key exceeds 6; the displayed key attains it. The mathematical work is construction *and* upper bound. If reading the full table is a barrier, arrange the five letter cards into loops and enumerate the same seven cases physically.

Upper: P then Q is not Q then P

P swaps A/B; Q swaps B/C; both fix D/E. P then Q takes ABCDE to BACDE to **CABDE**. Q then P takes ABCDE to ACBDE to **BCADE**. At letter A alone: A goes to B then C in the first order, but A then B in the second. Thus the maps differ. The composite P then Q is the loop A–C–B–A with D/E fixed: its order is **3**, even though each factor has order 2. To undo it, apply Q then P. Disjoint swaps, such as A/B and D/E, commute: neither changes any input moved by the other. Their common composite has two 2-cycles and order 2.

Extra: the eight-letter optimum is 15

Use a 5-cycle on A–E and a 3-cycle on F–H. Their first common return is 15. Exhaustive upper bound by largest loop: 8 gives 8; 7 leaves one and gives 7; 6 leaves at most a 2-cycle and gives 6; 5 leaves three letters, giving at most $\text{lcm}(5, 3) = 15$; 4 leaves four, with largest possible LCM 12 from 4+3+1; if all loops have size at most 3 their LCM divides 6. Thus nothing exceeds 15. For nine letters, 5+4 gives **20**. For completeness: largest 9,8,7,6 give maxima 9,8,14,6; largest 5 leaves at most four and gives at most 20; largest 4 gives at most 12; all at most 3 gives at most 6. Adding a letter never decreases the maximum (fix the new letter) but need not strictly increase it: sizes 5 and 6 both have maximum 6. On six letters, partitions with largest 6,5,4,3,2,1 have maximal orders 6,5,4,6,2,1.

The exact research question

Undergraduate mathematics

A reversible substitution on n symbols is a permutation, an element of the symmetric group. A repeated key is a power of this permutation; its return time is its order. Disjoint cycles, least common multiples, inverses, and noncommuting products are standard abstract algebra. No formal group axioms are needed in the child-facing activity. The words “key” and “code” describe the entry point; these small deterministic substitutions are not modern secure encryption. **Thomas Judson**, *Abstract Algebra: Theory and Applications*, Chapter 5, §5.1 “Definitions and Notation,” especially disjoint-cycle examples and Theorem 5.9 on disjoint-cycle decomposition. See university-hosted textbook, printed pp. 59–63. Our explicit loop proof supports the activity independently of theorem numbering.

A direct research lineage

Marc Deléglise, Jean-Louis Nicolas, and Paul Zimmermann, *Landau’s function for one million billions* (2008 arXiv manuscript), §1, defines $g(n)$, the maximum order of a permutation of n symbols, and relates it to least common multiples of partitions. The paper develops algorithms for very large n : arXiv:0803.2160; paper, introduction pp. 1–3. The classroom’s $g(5)=6$ and $g(8)=15$ are *exact small instances of the same optimization problem*. We do not claim that finding these small values is new research. Efficient computation and large- n growth are the research directions. The extra page asks for a proved maximum, not a decorated encoding exercise.

Downloaded sources and pedagogical choices

Rozhkovskaya, *Math Circles for Elementary School Students*, Lesson 2, problems 2.2–2.8 (EPUB section 12), uses secret codes and shifts. We retain the physical key and short-message entry while replacing the core with arbitrary reversible substitutions and loop structure. This week’s keys need not be shifts; week 4 imposes a fixed circular order. *Math Circle by the Bay*, preface pp. viii–x, supports a deep theme accessible across levels, manipulatives, explanation, and spare challenges. Rozhkovskaya Lesson 3, “At the lesson” item 1, supports attempted solutions and explanations before tabular organization. Our exact keys, tables, staffing, and timing are new adaptations.

Prior use and verification

Lowell Handout 8, problem 8.1, already used divisibility for distributing apples among 3,4,5,6 guests. The LCM idea here is revisited through synchronized permutation loops, with new questions about maximal orders. No saved use log proves these particular machines were taught. Record actual instances as F03-K/M/U/X-v2 and note which keys or partitions children saw. `verify.py` checks every permutation for $n=4,5,6,8,9$ against the claimed maxima and verifies all message traces and compositions. It is a separate check of the finite results, not a substitute for the loop and partition proofs.

Rotations, orbits, and shuffles

Week 3 allowed arbitrary permutations. Week 4 fixes a circular order and studies shifts: what controls their cycles and return times? The extra changes addition to multiplication through a real card shuffle.

Materials and preparation

Print 3 K-1, 3 middle, 1 upper, 1 extra reserve, and this guide. Bring seven counters, pencils, scratch paper, and cards 0-7 (another eight for the sixteen-card extension). For the K six-picture extension, draw shapes on scraps. Hop counters on the printed fixed rings; no movable wheel or brad is needed. Draw strips on scratch paper for the optional proof.

Readiness

K-1: match shapes, count hops to three with help; no independent reading. Reason about repeated movement and impossible landings. **2-3:** A-J familiarity, count up to ten; read-aloud removes word-reading barriers. Explain disjoint routes and use a common key. **4-5:** count to eighteen for finite rings. The optional general proof additionally needs division/gcd and the adult-supported coprime lemma below. Distinguish an orbit from the whole permutation. **Extra:** track numbered positions and remainders; powers can be explained as repeated doubling. The core eight-card proof needs only 2,4,8 and remainder 1.

A shared 60-minute hour

0-10: play with counters and rings, trying hop lengths. **10-15:** define clockwise and count movements, not the starting place. Launch: “Can a step rule miss some places forever?” **15-35:** K compares one/two-step routes; middle tries the code clue; upper explores the 12-ring. Adult A anchors K; organizer visits middle and upper. **35-40:** movement/reset: stand, stretch, leave work ready to return to. **40-55:** continue orbit comparisons or an optional proof below; offer the shuffle as an alternative. **55-60:** share a route, obstruction, or explanation; tidy counters and cards. The twelve-row table is optional: inverse shifts share return times. Move to explanation when ready. An adult should discuss the upper child’s conjecture; the three pages are choices, not a timed test.

Hints and common confusions

Count spaces moved, not the starting label. Stop an orbit when its start returns; then choose an unused start. Ask how many spaces and whole laps were traveled. In shuffling, distinguish a card label from its current position.

Satisfying stops / optional proof

K-1 stop: compare one/two-hop routes and undo the hidden turn. **Optional:** explain missed shapes using separate loops.

2-3 stop: decode and compare +2/+3 routes. **Optional:** prove with alternate markings that +2 misses half, yet has an inverse.

4-5 stop: explain +4/+8 returns and loops on twelve places. **Optional:** prove the gcd formula with the adult strip scaffold; otherwise label the general rule a conjecture or supplied theorem.

Exact routes and the gcd rule

K-1

From circle, one hop visits triangle, square, star, circle (four turns); two hops alternate circle/square (two turns), never landing on triangle or star. A secret circle-to-square shift is two hops, so triangle becomes star, square becomes circle, and star becomes triangle. Two hops undo themselves. Three-hop turns follow circle, star, square, triangle, circle, visiting all four. On six positions, among steps 1 through 5, only 1 and 5 visit all six; 2 and 4 give two loops of length 3; 3 gives three loops of length 2.

Middle

A to D is +3 on A-J. GDG decodes to **DAD**; BAG encodes to **EDJ**. B becomes E, so the additional B-to-F clue contradicts the fixed-shift rule. Repeated +2 gives A,C,E,G,I,A and B,D,F,H,J,B. Marking alternate letters explains why the two routes cannot meet: adding two preserves that marking. Repeated +3 gives A,D,G,J,C,F,I,B,E,H,A. The steps visiting all ten are **1,3,7,9**. Any shift is reversible: +2 is undone by -2, or +8. Skipping some letters along one repeated orbit is not information loss.

Upper: key and complete classification

9 to 1 gives +4, which takes 2 to 6. Outputs 1,4,10 decode to **9,0,6**. Its four loops are (0,4,8), (1,5,9), (2,6,10), (3,7,11), each returning in three turns.

Shift k	0	1	2	3	4	5	6	7	8	9	10	11
Return time	1	12	6	4	3	12	2	12	3	4	6	12
Number of loops	12	1	2	3	4	1	6	1	4	3	2	1

Steps 1,5,7,11 visit all twelve. The zero rule is the identity and has first *positive* return time 1, not zero. For eighteen positions, +6 returns after 3; +5 after 18.

Why the formula works

For +8 on 12 places, return means $3 \mid 2t$. Draw three t-strips, totaling $3t$. If the first two strips total a multiple of 3, removing them leaves t a multiple of 3. Thus the first positive return is 3. This proves this case for every t . The optional general bridge is on page 4; do not treat a completed table as its proof.

From turning to shuffling

Upper: inverse and composition

On twelve places, +4 is undone by +8 (or -4). For every starting x , +4 followed by +7 gives $x+11$ modulo 12; reversing the order gives the same because integer addition commutes. Every shift is bijective, even if its orbit from 0 is a proper subset. The visit-everywhere condition is $\gcd(n, k) = 1$, not a condition for reversibility. This distinction is central: a substitution permutation can have several cycles.

Extra: eight cards, three shuffles

Rows, left to right with the leftmost card the top:

Shuffle	Row
0	0 1 2 3 4 5 6 7
1	0 4 1 5 2 6 3 7
2	0 2 4 6 1 3 5 7
3	0 1 2 3 4 5 6 7

The old-to-new position map is **0,2,4,6,1,3,5,7**. Its cycles are fixed 0, fixed 7, (1,2,4), and (3,6,5). Thus the whole shuffle returns after 3 and not earlier. This is also a direct week 3 loop argument. For old position x in the first half (0–3), the new position is $2x$. In the second half (4–7), it is $2(x - 4) + 1 = 2x - 7$. For x below 7 these are exactly $2x \bmod 7$. Position 7 stays 7, separately; it must not be merged with residue 0. Repetition gives $2^t x \bmod 7$. The powers 2,4,8 have remainders 2,4,1, so the first identity occurs at $t=3$. Tracking position 1 rules out earlier global returns, and remainder 1 fixes all positions 0–6.

Extra predictions

Six cards use modulus 5. Powers 2,4,8,16 give remainders 2,4,3,1, so the order is **4**. Sixteen cards use modulus 15: 2,4,8,16 give 2,4,8,1, again order **4**. In general an out-shuffle on $2m$ cards maps positions below $2m - 1$ by doubling modulo $2m - 1$, fixes the final position, and has the multiplicative order of 2 as return time (the two-card identity is a trivial edge case). More cards can therefore have the same or even a smaller return time.

A useful additional contrast

Random shuffling asks about a distribution over possible orders. This perfect shuffle is one precisely defined permutation repeated deterministically. Its return time says nothing by itself about how mixed a random shuffle is. Do not introduce chance unless it becomes a separate, explicitly changed rule.

Mathematical lineage and reuse

Optional adult bridge: coprime divisibility

If positive a, b are coprime and $a \mid bt$, compare base strips a, b with scaled strips at, bt . Repeatedly subtract the smaller base strip from the larger, doing the same to the scaled strips. Both scaled lengths stay multiples of a . The positive base sum decreases, so the process ends with equal lengths c . Common divisors are unchanged: a divisor of x, y also divides $x-y, y$, and conversely by adding. Thus $c=1$, since the original lengths were coprime. The scaled process ends at t , which is therefore a multiple of a . Conversely, $a \mid t$ plainly implies $a \mid bt$.

Try 5,3 first: $5-3=2$, then $3-2=1$. On scaled strips this leaves $3t - (5t - 3t) = t$. Equivalently, $2 \cdot 3t - 5t = t$; if $5 \mid 3t$, the remainder is a multiple of 5. Ask the child to explain each removal. This is additional readiness, not assumed knowledge of prime factors.

For $k>0$, set $d = \gcd(n, k)$, $n = da$, $k = db$. Return means $a \mid bt$; the lemma gives $a \mid t$, hence first return n/d . Every start has this length, so there are d loops. For $k=0$, every point is fixed. If only the 3-and-2 case was explained, record that proof and label the general rule conjectured or given as a theorem. Full strip prompts and the Week 9 application: `plans/coprime-divisibility-scaffold.md`.

Sources and boundaries

Judson, *Abstract Algebra: Theory and Applications*, Ch. 4, §4.1, Theorem 4.13, pp. 48–49, gives the cyclic order formula: textbook. Our rings are its additive version. **Diaconis, Graham, Kantor (1983)**, *The mathematics of perfect shuffles*, §2, Lemma 1, p. 177, proves the out-shuffle order via doubling: original paper. The extra proves this small return-time case and position rule, not the paper's shuffle-group classification. The plan has full research references. Rozhkovskaya, Lesson 2, problems 2.3–2.7, supplies the shift-cipher entry. *Math Circle by the Bay*, preface viii–x, supports manipulatives, changing pace, and reserves; Rozhkovskaya Lesson 8 records copying slowing the activity. Our rings, proof scaffolds, and hour are adaptations. Lowell Handout 6, problem 6.7, and Handout 7, problem 7.8, use a different card-ordering solitaire. Ask about recognition; do not assume this shuffle was taught.

Use history and checks

Record actual steps, ring sizes, and extensions used as F04-K/M/U/X-v2. Keep 26-letter Caesar messages as a later alternative, not extra arithmetic for early finishers. Future years can compare prime/composite ring sizes, affine maps, in-shuffles, or perfect-shuffle groups; avoid replaying the same small cases. `verify.py` checks all 12 shifts, the 10-letter routes, message examples, the 18-ring predictions, shuffle rows, position formulas, and first return times for 6, 8, 16 cards. The facilitator proofs establish why the patterns hold.

A city is a mathematical object

Question: How much information forces one arrangement? Children progress from a complete list to a uniqueness proof to a lower bound on clues. These are questions about permutations, Latin squares, and determining sets.

Preparation and prerequisites

Use snap cubes: six per youngest child, eighteen per middle child, forty for the upper child: **112 cubes**, plus a few for the launch. Prebuild towers if time is short. Numeral slips can record a second solution after the physical attempt. Print three K-1 packets, three middle packets, one upper, one optional extra, and this guide. Build on the tabletop or quick scrap-paper grids with one-inch cells. The smaller printed grids record solutions; they are not sized to hold every brand of cube.

K-1: compare heights/count to 3; no independent reading; recognize different orders. Middle: count to 3 and follow row/column rules; adult may read; distinguish an example from a complete list. Upper: count to 4; track constraints, cases, and uniqueness; no formal algebra. Extra: understand a 4-by-4 Latin square, coordinate labels, and a pigeonhole argument.

Launch

Let towers be handled first. Show 2,1,3 from the left. Ask which towers a tiny person sees; touch each visible tower. Walk to the other end. Only after this introduce the separate city rule: every row and column uses each height once. A clue counts visible towers, not the first tower's height or the number of cubes.

A shared hour with seven children and two adults

0-10: explore materials freely. 10-15: brief launch and rules. 15-35: main investigation. 35-40: movement/reset; preserve work. 40-55: continue or choose an optional proof. 55-60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

K-1 stop: Build all six skylines and describe their views. **Optional proof:** organize by first tower to show completeness; explain the impossible one-at-each-end view.

Middle stop: Build both top-clue cities and select one with the second clue. **Optional proof:** certify the list, then swap unseen rows/columns to rule out every one-clue puzzle.

Upper stop: Explain the complete five-clue case table. **Optional proof:** check all five deletion certificates and prove the three-clue alternative forces the same city. The computer's minimum claim is separate.

Complete lists and minimum clues

K-1: six skylines, not six unrelated puzzles

The complete list (left count, right count) is: 123 (3,1), 132 (2,2), 213 (2,1), 231 (2,2), 312 (1,2), 321 (1,3). Fix the first tower: each of three choices leaves two orders. This proves completeness without factorial notation. One visible tower from both ends is impossible with distinct heights: the tallest would have to occupy both ends. Both 132 and 231 show two from each end.

The two 2-by-2 cities are 12/21 and 21/12. The top-row left clue 2 chooses the former. If row/column rules are too much, keep investigating rows; a complete skyline list is already substantial mathematics.

Middle: exactly two cities, then exactly one

Top clue 3 forces first column 1,2,3. Top row is either 123 or 132. Column exclusions force the respective completions:

1	2	3
2	3	1
3	1	2

1	3	2
2	1	3
3	2	1

The right clue 2 on row 2 keeps the first. Deleting that clue leaves both. Deleting only the top clue permits 123/231/312 and 312/231/123 (swap rows 1 and 3).

A genuine minimum proof: No single visibility clue can determine an entire 3-by-3 Latin square. If it sees a row, swap the two other rows; if it sees a column, swap the two other columns. The clue stays true and the square changes. Two clues suffice in our example, so the minimum over all uniquely solvable 3-by-3 visibility puzzles is two.

Hints and evidence

K: fix the first tower and exchange the other two. Middle: point down the first column, then try both top rows. Ask what rules out another answer. Accept a complete physical case split; long written prose is unnecessary. If children only solve the city, ask for one clue deletion and an alternative, saving the minimum proof for conversation.

A human certificate for five clues

Upper: the unique solution

Both the opening and the five-clue puzzle have rows **1234/2341/3412/4123**. Here is a small exhaustive argument for five clues. Bottom row starts with 4 and has right visibility 2, so 3 must be last. Only 4123 and 4213 remain. Keep both: the bottom clues alone do not rule either out; row 3 and row 2 do that next.

Row 3 has two views of 2. Its six possible rows are **1423, 2143, 2413, 3142, 3241, 3412**. Cross out candidates conflicting with the bottom row. The surviving pairs and possible row 2s are:

Bottom row	Row 3	Row 2 with right view 2
4123	3241	none
4123	3412	2341
4213	3142	none

Check the final column by listing row permutations avoiding the two entries already in each column, then testing the right view. The top row is forced by missing column heights. This proves uniqueness without trusting a solver.

Deleting clues: witnesses

Deleting L3, L4, R2, R3, R4 gives **2, 2, 6, 12, 20** solutions respectively. L3 means the left clue on row 3. One alternative per deletion proves necessity:

Delete	Alternative city, top row first
L3	1234 / 3412 / 2341 / 4123
L4	1234 / 4123 / 3412 / 2341
R2	1324 / 2431 / 3142 / 4213
R3	1324 / 3142 / 2431 / 4213
R4	1234 / 2143 / 3412 / 4321

On optional upper page 3, check each certificate is Latin, keeps all four retained clues, and differs from the target. Restoring the clue before the next deletion keeps the five arguments separate. The five-clue set is **minimal**, but it is **not minimum**: left clues 4,3,2 on rows 1,2,3 uniquely determine this same target. First row is 1234. The rows with left view 3 are 1243,1324,1342,2134,2314,2341; all but 2341 repeat a column entry. With rows 1 and 2 fixed, the possible row 3s are 3412 and 4123; only 3412 has left view 2. The last row is forced.

The script exhaustively verifies that three is the minimum *for this target city among subsets of its 16 perimeter clues*. That computation is not a proof that every 4-by-4 city needs three. For children, the concrete three-clue alternative already disproves the claim that minimal means minimum.

Trades, bounds, and research

Extra: a lower bound with an escape move

Each bold block has form ab/ba . Exchanging a, b inside it preserves the multiset in each affected row and column. With at most three entry clues, one of four disjoint blocks has no clue. Switch it: a distinct Latin square keeps every given entry. Thus at least four *entry* clues are necessary for this square.

Hitting these four blocks is not sufficient. Clues at $(1,1), (1,3), (3,1), (3,3)$ leave rows 2 and 4 unclued; swapping those rows keeps all four clues. Any uniquely determining set must meet *every* trade. Conversely, the changed cells in any alternative completion form a trade; hitting every such trade rules out alternatives.

Keep clue types separate: Previous pages use outside visibility clues; the extra uses fixed entries. Bounds do not transfer automatically.

Where the mathematics goes

Undergraduate combinatorics studies permutations, Latin squares, constraint satisfaction, and hitting sets. Joy Morris's *Combinatorics*, chapter on Latin squares develops Latin-square structure; our visibility questions add a different constraint system.

Hatami and Qian, *Teaching dimension, VC dimension and critical sets for Latin squares* (2016), §1 and Theorem 1.2, study sets of *entries* determining a square and prove a quadratic lower bound for sufficiently large orders. Our trade argument is a small instance of the question, not their asymptotic theorem. The paper states a sharper conjecture; no current open-status claim is needed here.

Teaching sources and returning children

JRMF *Skyscrapers Teaching Guide*, PDF pp. 1–4: physical viewpoints, city rules, and reasoning; pp. 13–14: blank boards. Our clue instances and investigations extend that format. *Math Circle by the Bay*, printed pp. ix–x: manipulatives, extra challenges, and themes at different depths. Lowell Handout 2.1–2.2 seating constraints are related prior material, not evidence children have solved these cities.

Use F05-K-v2/M-v2/U-v2/X-v2. Prepared, not recorded as taught. Save trade classification, Latin transversals, and larger skyline recurrences for later years. Rebuild with `build.sh`; `plans/verify-week-05.py` checks the finite claims independently of the explanations.

Useful questions; provably few tests

Question: Can we design a method that works for every secret, and prove no shorter method can always work? This is decision-tree complexity with exact-position feedback.

Prepare and check the rules

Use two kinds of counters (mark R/B or distinct symbols), paper position strips, and three folded-paper screens. About 12 counters of each kind per group suffice if guesses are drawn as they are made; keep all records visible. A parent may write dictated guesses and scores. First score visible secret RBR against RRR (2), BRB (0), and RBR (3). Repeated colors are allowed. Give only the **total correct positions**; no wrong-place-color feedback.

Print three K-1 packets, three middle, one upper, one optional extra. K: no reading; match positions and count 0-2; eliminate impossible examples. Middle: counts to 3, optional R/B reading; compare one changed feature and complete cases. Upper: counts to 4; worst-case reasoning, symmetries, and branching. Extra: 0/1 notation and small equations; variables can be replaced by counters.

Count the same thing throughout

A **test** is a scored query. Once exactly one secret remains, naming it is free in this investigation. Original game rules require actually submitting the secret, potentially one extra turn. Our exact optima for 2, 3, 4 positions are **2, 3, 4 tests to identify**, not claims of those same turn limits for winning submission. Stop early if a score already identifies the secret.

A shared hour with seven children and two adults

0-10: explore materials freely. 10-15: brief launch and rules. 15-35: main investigation. 35-40: movement/reset; preserve work. 40-55: continue or choose an optional proof. 55-60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

K-1 stop: Build a complete two-test method. **Optional proof:** for any first test, build two secrets scoring one, so one test cannot always work.

Middle stop: Explain why the three-test method works. **Optional proof:** combine the one-B triple obstruction with color renaming to cover every first test. Without renaming, claim only the RRR-first obstruction.

Upper stop: Explain why four tests always suffice. **Optional proof:** defend the triple and four-candidate branches separately, then use symmetry to prove optimality. Choose one older group's proof conversation at a time.

Two and three places: complete proofs

K-1: elimination becomes a decision tree

RR scores 0 only for BB, 2 only for RR, and 1 for RB or BR. In the last branch test RB: scores 2 or 0 distinguish them. So two tests suffice. One cannot always work: for every first test, two secrets match in exactly one position. Build those counterexamples.

Middle: a three-test upper bound

Let baseline RRR score s , the number of Rs in the secret. Flipping one test position R to B changes its contribution from 1 to 0 if the secret there is R, or 0 to 1 if it is B. The new score is $s - 1$ or $s + 1$, never unchanged. Tests BRR and RBR reveal positions 1 and 2. Their number of Rs subtracted from s determines position 3. This procedure covers all eight secrets.

Middle: why two tests cannot guarantee success

After RRR scores 2, candidates are BRR, RBR, RRB. The complete second-test table is:

Test	BRR	RBR	RRB
RRR	2	2	2
RRB	1	1	3
RBR	1	3	1
RBB	0	2	2
BRR	3	1	1
BRB	2	0	2
BBR	2	2	0
BBB	1	1	1

Every row repeats a score; some answer leaves two candidates. Conceptually, the three secrets differ only in the position of one B. A test only reveals whether that B falls inside its B positions, giving at most two outcomes. For the numbered symmetry step, physically turn BRB into RRR by swapping R/B at places 1 and 3 in both secrets and queries. An equal pair stays equal and an unequal pair stays unequal. Every score is preserved. Do this at the B positions of any first query; it becomes RRR, so the lower bound covers all first queries and all adaptive methods.

Hints

Make candidate rows tangible. The table is optional once a child explains the one-B structure and two possible scores; further entries add no proof. Test a troublesome pair first. Ask what score a test would earn for each candidate. A test need not itself be a possible secret to reveal useful information.

Four tests: upper and lower bounds

Upper bound

RRRR and three separate single-position flips determine places 1–3 by score changes; the baseline R count determines place 4. Four tests suffice. Generally this uses n tests to identify an n -bit secret, at most $n + 1$ turns if winning submission is required.

A difficult answer that stays consistent

By independent color renaming, take the first test to be RRRR. Answer 2, leaving six two-B secrets. Classify the second test by its B count. Position permutations and complementing a test (replacing score s by $4 - s$) reduce all cases:

0 or 4 Bs: no information; six remain.

1 B (BRRR): scores 3 or 1 each leave three candidates. Take score 3: BBRR, BRBR, BRRB. Ask: *Which position is fixed? What is the only feature that changes?* They share the first B; the other B occupies one of three places. Any last test has a fixed first-position contribution and only two possible scores on the remaining one-B triple. It cannot separate all three. The score-1 branch has the same structure; 3-B tests are complementary.

2 Bs (BBRR): score 4 leaves BBRR, score 0 leaves RRBB, score 2 leaves BRBR, BRRB, RBBR, RBRB. Choose the four-candidate branch. For any last test with k Bs and a two-B secret, let j positions have B in both. The score is $2 - k + 2j$. There are at most three possible scores, so four candidates cannot all be separated.

Every second test has a consistent branch that needs more than one further test. Thus three tests cannot guarantee identification; four is optimal. The lower bound uses actual score structure, not merely five possible answers per test.

Facilitating a hard proof

Let the child physically sort six code cards into score piles. Ask them to make the largest pile small, then play the stubborn setter. Page 1 supplies a satisfying stop. Pages 2–3 are an optional adult-supported lower-bound investigation; the final-query triple argument is different from the four-candidate counting argument. If full optimality is too much, obtain a proof of the four-test method and one failed three-test plan. Record that boundary honestly.

Relationship to Week 3: Here the hidden object is an ordered binary string, queries may adapt, and the goal is a worst-case measurement optimum. Score structure and adversarial lower bounds deepen earlier identification ideas.

Five bits and a research landmark

Extra: decode and prove

The example is **10110**, with scores 2,2,2,0. Let w count secret 1s and s_0, s_1, s_2, s_3 be scores in printed order. Then $w = 5 - s_0$, and

$$x_4 + x_5 = \frac{s_1 - s_0 + 2}{2}, \quad x_3 + x_5 = \frac{s_2 - s_0 + 2}{2}, \quad x_2 + x_5 = \frac{s_3 - s_0 + 2}{2}.$$

Each 0-to-1 change gains one match for a secret 1 and loses one for 0. If any pair-sum is 0 or 2, it fixes x_5 , then x_2, x_3, x_4 ; w fixes x_1 . Otherwise all three sums are 1. If $x_5 = 0$, the middle bits are 111 and w is 3 or 4. If $x_5 = 1$, they are 000 and w is 1 or 2. The ranges do not overlap. So w fixes the case and x_1 .

This proves four *preselected* tests distinguish all 32 secrets. The page claims sufficiency, not an independently proved five-place optimum. Matching scores encode binary-hypercube distances; distinguishing landmarks form a resolving set.

Undergraduate and research connections

Decision trees, adversary arguments, Hamming distance, and information bounds belong to discrete mathematics and algorithm analysis. The small optima here have explicit upper and lower proofs.

Cáceres et al., *On the Metric Dimension of Cartesian Products of Graphs* (2005 preprint; 2007 publication), §2, PDF p. 3, gives exactly our four five-bit landmarks. §6, PDF p. 8, equates exact-position static Mastermind with resolving sets in Hamming graphs. This directly adapts a finite construction from a research paper, with our elementary decoding proof.

Knuth, *The Computer as Master Mind* (1976–77), gives a five-guess strategy for commercial four-place, six-color Mastermind; see Knuth's author page and strategy link. That game also has wrong-place-color feedback. Its result does not directly apply here. The shared idea is minimizing the worst surviving candidate set.

Teaching lineage and future use

JRMF *Crack the Code*, PDF pp. 2–4 supplies binary exact-position play and the $n + 1$ -turn challenge. The optimum/adversary tasks and extra deepen it. *Math Circle by the Bay*, pp. ix–x supports manipulatives, varied pace, and optional challenges.

Use F06-K-v2/M-v2/U-v2/X-v2; prepared, not taught. Save more colors, noise, and average-case analysis for later. `plans/verify-week-06.py` searches all adaptive query trees for small optima and checks all 32 extra signatures.

Games are recursive mathematics

Question: Which player has a strategy against every reply? Can a finite chart justify an infinite pattern? What information is needed when two games are combined?

Prepare and launch

Bring 25 counters per group, large move cards **1 or 2**, **1, 3, 4**, and **1, 3, 5**, and pencils. Print three K-1 packets, three middle, one upper, one optional extra. Pre-numbered charts reduce copying. Two distinct mats identify the extra's two piles. For the memory reserve, bring a paper strip and a movable four-cell frame (or two folded scraps).

Let children move and count counters before rules. Demonstrate 1-or-2 from three counters; let a child move and check legality. Remove an allowed positive number no larger than the pile. Taking the last counter wins. Older children then receive the clearly different 1,3,4 card. Do not announce the losing-number pattern at the launch.

K: count/remove 1 or 2 and alternate turns; no reading; examine both replies. Middle: counts and subtraction to 14 (physical allowed); adult can read; distinguish one winning move from all losing options. Upper: subtraction/remainders to 28, with groups of seven as a concrete alternative; two-direction argument and induction. Extra: small nonnegative labels, smallest-missing operation, and decreasing two-pile games; no binary arithmetic needed for the printed proof.

A shared hour with seven children and two adults

0–10: explore materials freely. 10–15: brief launch and rules. 15–35: main investigation. 35–40: movement/reset; preserve work. 40–55: continue or choose an optional proof. 55–60: share and tidy. This flexible default is our proposal, not a source prescription.

One adult stays with K,K,1. The organizer alternates middle and upper, leaving a concrete next attempt and returning for an explanation. In games keep two opposing sides and rotate a checker. Proof continuations need adult attention; pages are not a completion quota.

K-1 stop: Demonstrate both replies that make three a trap. **Optional proof:** group counters in threes and explain the strategy for all multiples of three.

Middle stop: Defend W/L labels through seven. **Optional proof:** refute greedy play at eight and justify both winning moves at ten.

Upper stop: Conjecture losing remainders 0,2 and defend sample labels. **Optional proof:** show both remainder obligations and termination; reserve the paper-window argument for a later step.

Endgames build the whole strategy

K-1: groups of three

From 3, go second: answer 1 with 2 or 2 with 1. From 6, do the same, leaving 3; repeat. From 7 take 1 to leave 6; from 8 take 2; from 9 go second. Multiples of 3 are losing for the next player. Each pair of turns removes one group of three; the second player takes the last. Two demonstrated replies are a proof without a formula.

Middle: backward reasoning

For moves 1,3,4, positions 0–20 are:

Pile	0	1	2	3	4	5	6
W/L	L	W	L	W	W	W	W
Pile	7	8	9	10	11	12	13
W/L	L	W	L	W	W	W	W
Pile	14	15	16	17	18	19	20
W/L	L	W	L	W	W	W	W

Zero is L for the player who would move next. If a position has a move to L, take it: the position is W. If all moves go to W, each gives the opponent a winning strategy: it is L. Every move decreases the pile, so this bottoms out. It is backward induction, not a tally of who happened to win.

From 5 take 3 to leave 2; from 6 take 4. From 7 all moves leave 6,4,3, which are W. Alex's move from 8 leaves 5, and the opponent can take 3 to leave 2. Take 1 instead, leaving 7. From 10, taking 1 (leave 9) or 3 (leave 7) wins. Bea's uniqueness claim is false; taking 4 leaves W position 6.

Hints that preserve investigation

Ask about one or two counters. For a disputed label, lay out all legal next piles. Check whose turn the label describes. If a child calls 7 winning because they won once, replay as the opponent with correct replies; return the discovery to the child.

Returns: Lowell Handout 2.3 and Handout 3.1–3.3 used 1-through-3 and 1-through-5. Those establish previous materials, not mastery. Handout 3.1 mistakenly calls 5 losing after observing it can move to losing 4; with those old rules, 5 is winning. Use the checked definitions here.

An infinite proof from finite structure

Upper: losing remainders 0 and 2 modulo 7

Proposed L piles are $7k$ and $7k + 2$, including zero. Subtraction of 1,3,4 never connects these remainders. Every other remainder has a legal move into the set:

Remainder	1	3	4	5	6
Take	1	1	4	3	4
New remainder	0	2	0	2	2

Each move is legal even for the smallest positive representative. These are both proof obligations. Starting outside the set, move into it; every opposing reply exits; move back. The decreasing pile ends at zero, reached by your move. Starting inside gives the second player this strategy. The argument covers all piles, not just checked examples.

Change to 1,3,5

Every move is odd and changes parity. No move connects even piles. Every positive odd pile can take 1 to reach even. Even piles are L and odd piles W: second wins from 6 or 100, first from 7. The proof structure persists while the invariant changes. This is different from taking any number 1 through 5.

Why finite subtraction games eventually repeat

Copy labels onto a straight strip and frame four consecutive entries. To add the next label, inspect entries one, three, and four places back: the rightmost, second-from-left, and leftmost cells. Slide the frame one cell right after adding it. Match two windows and physically extend both to see the same next label. For 1,3,4, once four labels exist, the next depends only on those four. There are at most $2^4 = 16$ W/L windows. Among 17 windows some pair agrees. Identical windows give identical next labels; induction forces all future labels to repeat. This proves *eventual* periodicity, not repetition from zero or period seven. The remainder proof supplies those stronger facts here.

For any finite move set with maximum m , use m -label windows, starting after enough entries exist to use the full recurrence. Grundy values (next page) are bounded by the number of allowed moves, so their windows also eventually repeat. Infinitely many allowed moves do not provide a fixed finite-memory argument.

Adult attention and optional depth

The oldest child should defend a disputed label and a strategy to an adult. If page 2 takes the whole session, save the finite-window argument and extra. Record which proof actually appeared; a completed chart alone is not understanding why a pattern continues.

Beyond W and L: combining games

Extra: smallest-missing labels

For 1,3,4, labels for piles 0–10 are **0,1,0,1,2,3,2,0,1,0,1**. Thus (1,1) loses; (1,4) wins by taking 1 from the second pile to leave (1,3). Both (1,3) and (4,6) lose; (5,6) wins by taking 1 from the first pile to leave (4,6). W/L status alone forgets information needed when games combine.

The two-pile proof: By the smallest-missing definition, no option from label a has label a , and every smaller nonnegative label occurs among its options. A move from equal labels changes exactly one, destroying equality. With unequal labels, change the larger to the smaller; the definition guarantees such a move. Both losing-position obligations hold. Total size decreases, and terminal (0,0) has equal labels. This is the two-component case of Sprague–Grundy theory. For three or more piles the rule is bitwise XOR.

Undergraduate and research lineage

Ferguson’s UCLA Math 167 course includes take-away, Nim, graph games, and sums. His *Game Theory*, Part I, §1.3–1.4 develops winning/losing and subtraction games; §3.2 defines Sprague–Grundy values and §4.2 treats sums. Our proofs are concrete instances.

Shenxing Zhang, *On the linearity of the periods of subtraction games* (2024), §1, Lemma 1.1 and Conjectures 1.2–1.3, investigates changes in periods and preperiods when one move is added to a fixed set; Theorem 1.4 proves the predicted behavior in several families. Our seven-period example is elementary; predicting whole families is research. We describe the paper’s conjectures, not an exhaustive survey of their current status.

Sources, reserves, and verification

JRMF *Countdown Teaching Guide*, PDF pp. 1–2: play-first questions; pp. 4–5: W/L reasoning; p. 8, challenge 4 explicitly uses 1,3,4. Our parity comparison, finite-window scaffold, and extra extend it. *Math Circle by the Bay*, printed pp. ix–x supports manipulatives and local extensions; the two-adult arrangement is ours.

Use F07-K-v2/M-v2/U-v2/X-v2, prepared, not taught. Save misère play, other move sets, and three-pile XOR for later. `plans/verify-week-07.py` checks 1,001 single-pile values and independently solves two-pile games with each pile at most 30; proofs above establish the unbounded claims.

From mirror play to Nim

Core question: Which positions can the next player force a win from, and why? Children first play; classifications must describe perfect opposition, not the result of one game. All games use normal play: taking the last counter wins.

Prepare for seven children and two adults

Print 3 K-1 packets, 3 middle packets, 1 upper packet, and keep one extra page in reserve. Bring about 60 counters, five paper plates or marked pile areas for the middle/upper table, pencils, and scraps labeled 1, 2, 4, 8. Reuse counters between trials. The boards need no calibrated size.

K-1: no reading; compare small distances and count to 2. Middle: short instructions read aloud, counts to 7, reasoning about all replies. Upper: subtract small integers and distinguish odd/even; binary notation is taught through bundles. Extra: organized lists and subtraction; floor/square-root notation is explained, not a prerequisite for the first two questions.

A flexible shared hour

0-10: freely move tokens and build piles. 10-15: briefly demonstrate legal long/short right or down moves and an illegal diagonal. 15-35: main investigation: parent starts K-1; organizer launches the middle board and three piles, then alternates between them about every five minutes. Leave a specific start to test. 35-40: movement/reset. 40-55: continue play and explanations, or choose a proof continuation. 55-60: share a move with its reason and tidy.

Triplets rotate two opponents and a checker; the checker does not become a third player. Parent remains mainly with K-1. The fifth grader gets two actual adult games and a proof conversation; a spare worksheet does not replace an opponent.

Launch and hints, in order

“Can you leave me a square where every move helps you?” First ask children to show every legal move. Then ask which distances remain to the star. Offer matching only after exploration. Upper: try small counterexamples before giving bundle cards; ask “Which bundle is the largest unpaired one?” Stop after a satisfying explanation; finishing all pages is not required.

A satisfying stop; an optional proof

K-1: show both center moves and their replies; continue by explaining copying from any equal piles. **Middle:** map the board and restore equality; continue with both directions of the arbitrary two-pile proof and termination. **Upper:** refute even-total reasoning and find a legal bundle-balancing move; continue with the full binary proof. Record an explained claim separately from a theorem supplied by an adult.

Checked two-pile reasoning

K-1 and middle board answers

Distances (a, b) mean squares remaining right and down. A rook move reduces exactly one positive coordinate. The star is $(0, 0)$: the previous player has already won. On the 3×3 board, the center $(1, 1)$ is losing. Both possible moves allow an immediate reply to the star. Top-left $(2, 2)$ is also losing; copy the amount removed in the other direction.

On the 4×4 board, the traps form the top-left to bottom-right diagonal: $(3, 3), (2, 2), (1, 1), (0, 0)$. Every unequal pair is winning: reduce the larger coordinate to the smaller. The middle start $(3, 2)$ has the unique winning move to $(2, 2)$.

A complete proof, with counters

If two piles have the same size, every move makes them unequal. If they differ, reducing the larger to the smaller is legal and restores equality. Thus a player who hands over equal piles can always restore equality after an opponent's move. The total number of counters strictly decreases, so play must finish. The player restoring equality takes the final counters; the opponent receives $(0, 0)$ and cannot move. These two properties establish all traps, on rectangles as well as squares.

Any unequal two-pile start has exactly one winning first move (specifying the pile and amount). Matching piles have none. Children can use $(5, 2)$ as their designed puzzle: remove 3 from the 5-pile.

Why three piles change the problem

$(1, 1, 1)$ is winning: remove one pile and leave $(0, 1, 1)$. Even total is insufficient: $(1, 1, 2)$ has total 4 but is winning by removing the whole 2-pile. $(1, 2, 3)$ is losing. Here are all first moves and one reply; pile order is kept fixed.

After move	Reply leaves	After move	Reply leaves
$(0, 2, 3)$	$(0, 2, 2)$	$(1, 2, 0)$	$(1, 1, 0)$
$(1, 0, 3)$	$(1, 0, 1)$	$(1, 2, 1)$	$(1, 0, 1)$
$(1, 1, 3)$	$(1, 1, 0)$	$(1, 2, 2)$	$(0, 2, 2)$

Each reply leaves two equal piles and one empty pile. This gives a physical proof before binary notation. The middle group can stop here or join the upper bundle experiment.

The binary invariant

Upper answers

Binary records: $1 = 001$, $2 = 010$, $3 = 011$, $4 = 100$, $5 = 101$, $6 = 110$, $7 = 111$. $(1, 2, 3)$ and $(2, 4, 6)$ have even column sums. $(1, 2, 4)$ does not; reduce 4 to 3. For the table: $(2, 4, 6)$ is balanced; $(3, 4, 5)$ can become $(1, 4, 5)$; $(1, 5, 7)$ can become $(1, 5, 4)$; $(3, 5, 6)$ is balanced. Other correct balancing moves should be accepted.

For $(7, 10, 12)$, the Nim sum is 1; reduce 7 to 6. For chosen first piles a, b , the unique third pile is $a \oplus b$: each binary digit is forced by whether the first two digits agree.

Why the test is sufficient, for any number of piles

A move changes exactly one integer. At least one of its binary digits changes, so at least one previously even column becomes odd. No legal move can connect two balanced positions.

If the Nim sum s is nonzero, choose its highest 1-bit, say the 2^k column. Some pile x has a 1 there. Replace it by $y = x \oplus s$. Higher bits stay fixed, this bit becomes 0, and lower bits may change. The sum of all smaller bundles is $2^k - 1$, so $y < x$: the move is legal. Every odd column is toggled once, every even column is unchanged, and the result is balanced. The decreasing total ensures termination. These two facts prove exactly the same win/lose structure as in the two-pile proof.

Do not physically permit removing an arbitrary bundle while leaving a larger number of counters: the game is still ordinary removal from one pile. The bundle display is a way to calculate a legal smaller result.

Extra: Wythoff, checked results and proof boundary

The old equal-pile trap fails because both piles can be emptied in one diagonal move. For $0 \leq a \leq b \leq 7$, the traps are $(0, 0)$, $(1, 2)$, $(3, 5)$, $(4, 7)$. Greedy pairs through $n = 5$ are $(0, 0)$, $(1, 2)$, $(3, 5)$, $(4, 7)$, $(6, 10)$, $(8, 13)$.

Distinct pairs have disjoint coordinates and distinct absolute differences, so neither a single-pile move nor an equal-take move connects them, including reversed pile order. This proves only the no-move-between-traps condition. To prove completeness one must also show every outside position reaches a listed pair. Sample moves: $(2, 4) \rightarrow (2, 1)$, $(5, 8) \rightarrow (4, 7)$, $(8, 12) \rightarrow (6, 10)$. The golden-ratio formula is a cited theorem, not a conclusion from five data points.

Where the mathematics leads

Undergraduate direction and genuine research

Two piles give a strategy-preserving change of representation from a grid to an impartial game. Binary column parity is addition in a vector space over the two-element field. The proof has a useful algorithmic form: recognize the losing set, prove no move stays inside it, and construct a move into it from everywhere else. Formal algebra vocabulary can wait.

Charles Bouton's historical paper, *Nim, a Game with a Complete Mathematical Theory*, *Annals of Mathematics* 3 (1901–02), pp. 35–39, is the original research source for binary analysis. Our normal-play proof is written out on page 3, so the circle does not depend on reading the historical notation. Caltech-hosted paper.

Fraenkel and Peled, *Harnessing the unwieldy MEX function*, *Games of No Chance 4*, MSRI Publications 63 (2015), pp. 77–94: §1, pp. 77–78 states the Wythoff rule, greedy pairs, and golden-ratio formula; §5 develops efficient computation for generalized complementary sequences. Author research chapter.

The extra page meets an actual research theme: a simple rule change can replace bitwise arithmetic with complementary sequences and a question of efficient computation. It does not reproduce the paper's generalized algorithm or claim an open problem has been solved by the classroom experiment.

Local sources and teaching choices

JRMF, *Rook's Move Activity Guide*, PDF pp. 3–6: legal moves, exploration before advice, losing positions, and the Wythoff connection. We replace the old upper king variant with three-pile Nim; the king version remains an untaught reserve unless the use log says otherwise.

Math Circle by the Bay, preface printed pp. viii–x (PDF pp. 9–11), supports common themes with adjustable depth, manipulatives, explanations, and extra challenges. Rozhkovskaya, Lesson 7 “At the lesson,” EPUB part0017.xhtml, highlights checking that children understand legal moves. The seven-child schedule and bundle prompts are our adaptations.

Lowell Handout 2, problem 2.3, and Handout 3, problems 3.1–3.3, include subtraction-game history; Week 7 also studies constrained subtraction. Here an entire pile may be removed, and the important new step is multiple piles plus binary balance. Reuse is deliberate, not evidence that every child has seen these exact starts.

Record and verify

IDs F08-K/M/U/X-v2. Record the actual starts, rule, level, strategies explained, and whether Wythoff was attempted. Keep unused stages for later years. `verify.py` checks all three-pile states through size 12 and Wythoff states through 30 by independent backward recursion. Those finite checks support the examples; the proofs establish the unlimited claims.

From bouncing to straight lines

Core question: Predict the first corner and the number of bounces without tracing the whole path. Launch from bottom-left at 45 degrees. Stop at the FIRST corner; do not reflect there. Line crossings inside the table have no effect.

Preparation, access, and the shared hour

Print 3 K-1, 3 middle, 1 upper, and one extra reserve page. Bring seven counters, rulers, pencils, scrap grid paper, and two spare copies of the K-1 folding page. Folding is optional; an adult can reflect a traced room instead. Allow about 10 minutes to prepare; no actual ball is needed.

K-1: no reading or arithmetic; track a diagonal and choose one changed direction. Middle: count lengths and multiples to 12; coordinate two repeating lists. Upper: multiples, parity, and division; introduce gcd/lcm after a picture. The general coprime lemma additionally needs adult-supported subtraction of multiples and Euclid's argument. Extra: fractions and coordinate pairs; irrationality of $\sqrt{2}$ may be supplied as known, with its proof available on page 3.

0-10 explore counters and mirrored grid rooms. 10-15 brief reflection launch: which direction is blocked by a right wall? 15-35 main investigation: parent models K-1 while organizer launches middle and upper, then alternates visits; children trace, fold, and predict. 35-40 movement/reset. 40-55 continue with target designs or an optional proof. 55-60 share a prediction with its reason and tidy.

Triplets rotate tracer, predictor, and checker but each child should move a token. Parent mostly stays with K-1. The fifth grader receives a personal discussion, with a concrete next table left between visits.

Launch and hints

"What if we opened the wall into a mirror room and kept going straight?" Let the child fold a drawn path before introducing number theory. Ask which copied walls it reaches together. If tracing is difficult, adult draws while child chooses directions. Upper hint order: compare a rectangle with a scaled copy; list common multiples; count whole rooms; use odd/even; only then reduce by gcd.

For returners, ask for one remembered corner prediction, then go straight to mirrored rooms and explain it. Skip familiar traces and redundant table rows; the new destination is unfolding, an obstruction, or inverse design.

A satisfying stop; an optional proof

K-1: predict and fold one route; continue by explaining the different 2-by-4 and 4-by-2 corners. **Middle:** build and explain a target table; continue by justifying the first common wall multiple. **Upper:** rule out bottom-left by an earlier corner and count walls; continue with adult-supported coprime division and the complete inverse design. Do not require every table row.

Checked paths and corner rule

K-1 and middle answers

For width w , height h : 2×2 ends top-right with no bounce. 2×4 follows $(0, 0)$, $(2, 2)$, $(0, 4)$, ending top-left; 4×2 follows $(0, 0)$, $(2, 2)$, $(4, 0)$, ending bottom-right. Folding the two width-2, height-4 rooms turns the straight diagonal through $(2, 2)$ into the first of these paths.

For 2×3 , the folded path is $(0, 0)$, $(2, 2)$, $(1, 3)$, $(0, 2)$, $(2, 0)$. In copied rooms, the first simultaneous wall meeting is $(6, 6)$. The four middle rows have $(L, \text{widths}, \text{heights}, \text{corner})$:

Table	L	Widths	Heights	Corner
2×3	6	3	2	bottom-right
3×6	6	2	1	top-left
3×9	9	3	1	top-right
4×6	12	3	2	bottom-right

A middle design reaching top-right after a bounce can be 2×6 (two bounces); top-left can be 2×4 . Accept any verified design.

Upper theorem with all hypotheses

Take positive integer w, h and a 45-degree launch. Reflections unfold to $y = x$. A corner occurs when the common coordinate is both a multiple of w and of h . Its first positive value is $L = \text{lcm}(w, h)$. First use room counts $m = L/w, n = L/h$. If they shared $d > 1$, distance L/d would already cross m/d widths and n/d heights: an earlier corner. Thus they are coprime; both cannot be even, ruling out bottom-left without a gcd formula.

Optional arithmetic bridge. Write $w = ga, h = gb$ with coprime a, b . Any common multiple is gat with $b \mid at$. To justify $b \mid t$, begin with $a = 2, b = 3$: both $3t$ and $2t$ divide into threes, so their difference t does too. In general, run repeated subtraction on strips of lengths a, b and, alongside them, at, bt . Subtract the shorter from the longer until equal. Common divisors are unchanged, so the coprime base strips end at 1 and the scaled strips at t . Both scaled lengths remain multiples of b throughout; hence $b \mid t$. The least positive t is therefore b , and $L = gab, m = b, n = a$. The shared plan scaffold supplies physical bundle examples. If this step is supplied, record it as a given theorem.

An odd number of whole widths folds to the right, even to the left. An odd number of whole heights folds to the top, even to the bottom. Thus odd a , odd b gives top-right; even a , odd b gives bottom-right; odd a , even b gives top-left. Both cannot be even, so first arrival at bottom-left is impossible. Scaling both dimensions preserves the reduced pair and corner.

Upper rows: 4×6 : $L = 12, m = 3, n = 2$, bottom-right; 6×9 : 18, 3, 2, bottom-right; 6×10 : 30, 5, 3, top-right; 8×14 : 56, 7, 4, bottom-right. Raw parity of the original lengths does not determine the endpoint.

Bounces, inverse design, and slopes

Count crossings, then design

Before the final corner the line crosses $m - 1$ vertical room walls and $n - 1$ horizontal ones. With the optional lemma these are $b - 1$ and $a - 1$. No crossing belongs to both lists: that would be an earlier corner. So the number of bounces is $a + b - 2$ (not counting launch or terminal corner). Path length, if wanted, is $\sqrt{2}L$; L itself is horizontal distance, not path length.

For top-right after exactly four bounces, a, b are coprime odd positives with $a + b = 6$. Candidates $(1, 5), (3, 3), (5, 1)$ reduce to just $(1, 5), (5, 1)$ because $(3, 3)$ is not coprime. Any positive integer scaling works, such as 2×10 . Top-right after three bounces is impossible: two odd numbers have even sum, so $a + b - 2$ is even. The latter is a parity obstruction, not a failed search.

Extra page: rational and irrational directions

The extra uses a *unit square* and physical slope $s = p/q$ in lowest terms. A positive lattice point (u, v) on $y = (p/q)x$ satisfies $qv = pu$, so its first occurrence is (q, p) . By the coprime lemma on page 2, q divides u ; the first positive choice is $u = q$, giving $v = p$. The endpoint's horizontal side is right iff q is odd; its vertical side is top iff p is odd. There are $p + q - 2$ bounces. Slope $2/3$ ends bottom-right with 3 bounces; $3/2$ ends top-left with 3.

For slope $\sqrt{2}$, an integer corner (u, v) with $u > 0$ would give $\sqrt{2} = v/u$, impossible. If requested, prove irrationality: reduce a supposed fraction v/u first. From $v^2 = 2u^2$, v is even; substituting $v = 2k$ forces u even, contradicting the reduction.

The other launch $(0, 1/2)$ with slope 1 cycles through $(1/2, 1), (1, 1/2), (1/2, 0), (0, 1/2)$. It avoids corners but follows just a diamond. No-corner does not imply dense. Density of the $\sqrt{2}$ -slope trajectory launched from $(0, 0)$ is a separate theorem. Under our stopping rule, arbitrary starts need care: from $(0, 2 - \sqrt{2})$ the same slope reaches unfolded $(1, 2)$, folds to bottom-right, and stops.

Keep the conventions straight

A general rectangle with physical slope s becomes a square after rescaling axes, with normalized slope sw/h . Rationality of that normalized slope controls the square model. The extra fixes $w = h = 1$ to avoid silently mixing those slopes. "Rational polygon" in research means its angles are rational multiples of π ; it does not mean a rational launch slope. Our corner-start paths stop when they hit a corner, so do not call those stopped paths periodic billiard orbits.

A doorway into dynamics

Mathematical direction and primary sources

Unfolding replaces a piecewise reflected path by a straight line on a lattice. The corner problem is divisibility, coprime integers, and parity; the inverse problem asks for a complete classification of parameters. These are substantive undergraduate number-theoretic and geometric ideas in a tangible representation.

Masur and Tabachnikov, *Rational billiards and flat structures*, published in *Handbook of Dynamical Systems* 1A (2002), pp. 1015–1089. In the linked author manuscript, §1.3 gives unfolding, §1.4 develops the unit-square/torus model, and §1.5 extends the construction to flat surfaces. Later §4 concerns periodic orbits. Author manuscript.

The classroom genuinely performs the first construction used in the research: reflect copies, then study straight motion. It does not establish the deeper recurrence or ergodicity theorems. The extra explicitly separates avoiding corners from density; the latter is supplied specifically for the origin-start $\sqrt{2}$ path under our stopping convention.

Downloaded sources and documented prior use

JRMF *Billiards Geometry*, PDF p. 2, asks for corner, bounce-count, and path-length rules. Lowell Handout 10, problems 10.2–10.7, includes 7×5 , 4×3 , 8×10 , scaling, consecutive sides, $1 \times n$, and $2 \times \text{odd}$ families. Handout 11, problems 11.4–11.5, includes height-five rectangles and plus-shaped tables. Do not describe small rectangle families or scaling as wholly new for returning children.

Our changed emphasis is a proof through physical unfolding, an exact first-corner criterion, a bounce formula, and inverse existence/impossibility design. This deliberately promotes an old reserve into the core. Plus-shaped tables remain a later branch; do not claim the rectangle formula covers them.

Math Circle by the Bay, preface printed pp. viii–x, supports manipulatives and revisiting ideas at different depths. Rozhkovskaya, Lesson 3 “At the lesson,” EPUB part0013.xhtml, supports attempts and child explanations before formal displays; Lesson 8’s account warns about copying pace. Preprinted grids and optional adult tracing are our response, not a procedure claimed to be in those sources.

Keep a useful history

IDs F09-K/M/U/X-v2. Record actual dimensions, launch slopes, whether unfolding was used, which formulas were explained, and which claims stayed conjectural. A prepared extension is not a taught activity. Future variants can change slopes, starting points, or polygons instead of enlarging the same arithmetic table.

`verify.py` independently traces all 900 rectangles with sides 1–30, compares the corner and bounce formulas, and checks the inverse-design list. The general proofs are on pages 2–3; finite checks alone do not establish them.

Puzzle makers become route planners

Core question: When is an every-road-once walk possible, and when repetition is necessary, how little can we repeat? Begin by moving a counter and marking used roads. Every drawing has a connected network and undirected roads. Each road joins two different dots; parallel roads are allowed.

Preparation and access

Print 3 K-1, 3 middle, 1 upper, and one extra reserve page. Bring seven moving counters, about 35 small counters to mark used edges, pencils, scrap paper, and two short paper strips for the upper loop splice. Strings between paper islands are an optional larger model. Setup takes about 10 minutes. No ruler accuracy or actual bridge construction is required.

K-1: no reading; distinguish a line from its endpoints, keep track of used bridges, reason about dead ends. Middle: count degrees to 4 and recognize odd/even; understand that failure of a few attempts is not proof. Upper: follow a route list, count edges, give a general argument with paired arrivals/departures. Extra: add lengths to 23, compare all three pairings, follow a shortest-path and lower-bound argument; no algebra beyond these small sums.

A shared hour with a meaningful workshop

0–10 build little maps and walk freely. 10–15 briefly demonstrate every-bridge-once, allowing island revisits. 15–35 main investigation: parent starts K-1; organizer launches the house and two-triangle splice, then alternates visits about every five minutes. 35–40 movement/reset. 40–55 continue repairs or shortest delivery, or choose a proof continuation. 55–60 share a reason and tidy.

Triplets rotate walker, edge-marker, and skeptic. Parent stays mostly with K-1; organizer visits middle and upper about every five minutes. The fifth grader should test a conjecture against an adult and explain a lower bound, not merely receive harder numbers.

Launch and hint sequence

“Can you cross every bridge once without teleporting?” Initially leave start and finish free. Later ask “What must happen after you arrive at a place in the middle?” Pair incident edges with fingers. Reveal odd/even vocabulary after the physical pairing. For route optimization ask why repeating one road changes parity at only two places. Offer the three possible pairings only after a first route attempt.

The puzzle-making ending has a concrete specification, a construction, and a certificate. Archive the route as well as the picture; testing by one partner alone does not prove optimality or impossibility.

A satisfying stop; an optional proof

K-1: explain the three-leaf obstruction and show a repair; continue by forcing its endpoints. **Middle:** explain house endpoints and repair K4; continue with the general odd-endpoint argument. **Upper:** splice the two triangle loops, then attain and certify delivery length 8; continue with the all-even construction and temporary-edge proof. Record only the claims actually explained.

One-walk maps: checks and proof

K-1 answers

A triangle has a closed route, e.g. $A-B-C-A$. Add a branch $A-D$: a trail is $D-A-B-C-A$, with endpoints D and A . At a degree-1 leaf, a walk cannot arrive and then leave by a different unused bridge. Thus the three-leaf star cannot have a trail covering every edge once: it would need three endpoints, while a walk has at most two.

Adding a bridge $B-C$ between two leaves repairs the star: $D-A-B-C-A$ is a complete trail. It begins/ends at the remaining leaf D and center A . Accept another pair of leaves with relabeled route.

Middle and upper map answers

House degrees: $A=2, B=2, C=3, D=3, E=2$. One trail is $D-A-B-C-D-E-C$. Every complete trail must start at one of C, D and end at the other. No closed Euler trail exists.

For the triangle with center D , all four vertices A, B, C, D have degree 3. It has no every-edge-once trail. Add a second AB bridge; now A, B are even and C, D odd. A valid trail is $C-A-B-D-A-B-C-D$, using the two distinct AB bridges separately. Middle children may draw the added bridge as a curve; a crossing without a dot is not a new junction.

Euler criterion with a construction

In any completed trail, arrivals and departures at an internal vertex pair. A closed Euler trail therefore has only even degrees; an open Euler trail has exactly two odd vertices, its endpoints. The necessity argument includes repeated visits to a vertex: each visit contributes another pair.

First physically splice the two triangles ABC and ADE sharing A . Mark AB, BC, CA used after $A-B-C-A$; AD, DE, EA remain. Insert the strip $A-D-E-A$ at a visit to A to obtain, for example, $A-D-E-A-B-C-A$. Trace it and mark all six roads. This one construction motivates the next general argument.

For sufficiency, assume a finite connected graph with at least one edge and all degrees even. Follow unused edges until stuck. At a vertex other than the start, arriving uses one edge of an even available count and leaves at least one to depart; so only the start can be where the walk stops. If unused edges remain, connectedness ensures that some vertex of the current closed walk touches unused edges (take a path toward an unused edge). Start a new closed walk there in the remaining even-degree graph and splice it into the first. Repeat. Finitely many edges guarantee completion.

For exactly two odd vertices, add one temporary edge between them, construct the closed Euler trail, then remove that temporary edge to obtain an open Euler trail. Parallel edges are allowed. This proves existence; it does not promise that every arbitrary greedy walk uses all edges without splicing.

Delivery routes and optimality

Upper: six unit-length roads

The triangle-and-center network is K_4 , with six unit roads. Every closed covering walk must repeat at least two road crossings: all four original vertices are odd, and each added traversal changes parity at its two endpoints only. A closed walk has even traversal-degree everywhere. Thus six original plus at least two repeats gives a lower bound of 8.

A matching route is A–B–C–D–A–C–D–B–A. It repeats AB and CD; all other roads occur once. The three possible odd-vertex pairings are AB+CD, AC+BD, AD+BC. Each adds two roads and creates an even connected multigraph, so each admits a closed Euler trail of length 8.

If finishing anywhere is allowed, at most two odd vertices may remain. At least one repeated crossing is necessary, giving a lower bound of 7. A–B–C–A–D–C–D–B attains 7, repeating CD.

For the design brief, a star with five leaves has six odd vertices: five degree-1 leaves and the degree-5 center. But its leaves are not adjacent, so it cannot meet the three-extra-step target. A workable design is two triangles joined by three corresponding cross edges (a triangular prism graph). All six degrees are 3. Duplicate those three cross edges; now all degrees are 4, so a closed tour exists with exactly three extra crossings, meeting the parity lower bound. Label the triangles ABC and DEF, with cross edges AD, BE, CF. One 12-step closed route is A–B–C–A–D–E–B–E–F–C–F–D–A. Children may find other designs.

Extra: six roads of different lengths

Original total is $2 + 3 + 7 + 1 + 6 + 1 = 20$. Distances between odd vertices are $d(A, B) = 2$, $d(A, C) = 2$ via D, $d(A, D) = 1$, $d(B, C) = 3$, $d(B, D) = 3$ via A, $d(C, D) = 1$.

Pair costs: AB+CD=3; AC+BD=5; AD+BC=4. Choosing the cheapest available AD first is not guaranteed optimal. Duplicate AB and CD to obtain minimum total 23. The same A–B–C–D–A–C–D–B–A route has length 23. Long roads AC and BD must still be serviced once; shortest connections are used only for the *extra* travel.

For a general lower bound, subtract one required copy of every road from a closed covering walk. The remaining multigraph has odd degree exactly at the original odd vertices. Its edges can be decomposed into paths pairing those odd vertices and closed loops: follow unused edges from an odd vertex to another odd one, remove that path, and continue, then remove cycles. Each path costs at least the shortest distance of its endpoint pair, and loops cannot improve a nonnegative-cost total. Thus every solution pays at least the best pairing cost. Duplicating shortest paths for that pairing attains the bound by Euler's construction.

The mathematics behind the workshop

Undergraduate graph theory

The optional proof conversation completes the Euler trail criterion, including its constructive direction; the physical splice is a satisfying earlier step. It distinguishes feasibility from optimization and separates an example route (an upper bound) from an obstruction (a lower bound). The upper activity turns parity into an algorithm for a small route-inspection problem.

Cornell CS 2800, Spring 2017, *Lecture 36: Paths*, gives the even-degree theorem and a cycle-splicing proof. Official course notes. Our maps, graded questions, and counter-based launch are original adaptations.

A real research connection

Edmonds and Johnson, *Matching, Euler tours and the Chinese postman*, *Mathematical Programming* 5 (1973), pp. 88–124, studies the weighted route-inspection problem using matching theory. The introduction describes the problem and its structure; §4 concerns a matching algorithm and §5 Euler tours. IBM publication record; Paper hosted by University of Michigan.

The extra page performs an exact four-odd-vertex instance of the same reduction. It does not teach the general blossom algorithm, polyhedral proof, or directed/mixed variants. For $2k$ odd vertices, exhaustive pairings number $(2k-1)(2k-3)\cdots 1$; efficient weighted matching makes large instances practical. The research is a historical solved result, not a claimed present-day open problem.

Local sources and returners

Rozhkovskaya, *Math Circles for Elementary School Students*, M.4.1 “Make your own problem,” EPUB part0031.xhtml, inspires making under constraints. Its animal/figure/number prompt is different; our connected-map/parity specifications are our design. Lesson 3 “At the lesson” supports child attempts and explanations before an organizing display.

Math Circle by the Bay, preface printed pp. viii–x, supports adjustable depth and extra challenges. Our timetable and two-adult allocation are proposals for this roster.

Lowell Handout 10’s password-window puzzle is related to directed Euler/de Bruijn constructions; do not call graph traversal wholly new. Here the representation is an undirected bridge map, and the claims concern odd-degree obstructions and minimum repeated travel. Week 1’s flip graph concerned states and distance; here edges are roads that must all be serviced. Generic old workshop starters remain optional future reserves.

Use history and checks

IDs F10-K/M/U/X-v2. Record maps, repairs, exact routes, whether optimality was justified, and child-designed problems with their solutions. `verify.py` independently searches states (current vertex, set of roads already visited) and confirms closed/open/weighted optima 8, 7, 23. It also checks printed routes. The mathematical certificates above explain why the results hold.